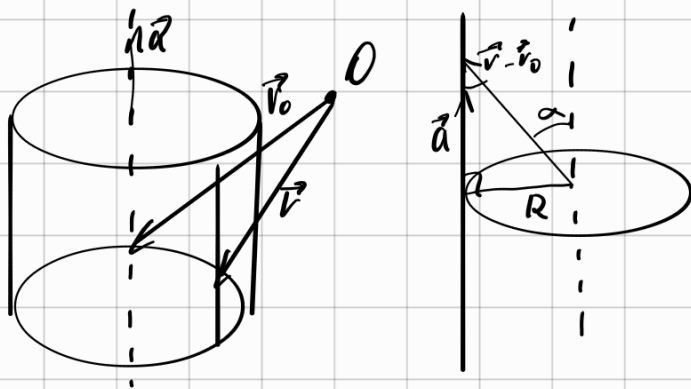


10.26 Рассмотрим вектор $\vec{r}$ в плоскости кругового цилиндра радиуса R с осью $\vec{r} = \vec{r}_0 + \vec{a}t$



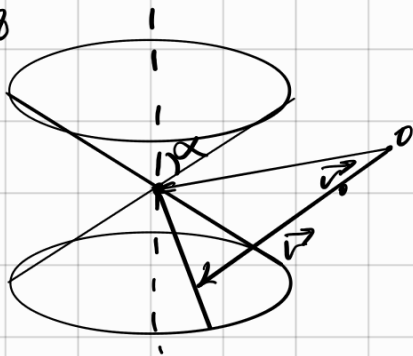
$$\frac{R}{|\vec{r} - \vec{r}_0|} = |\sin \alpha|$$

$$R = |\vec{r} - \vec{r}_0| |\sin \alpha|$$

$$R|\vec{a}| = |\vec{r} - \vec{r}_0| |\vec{a}| |\sin \alpha|$$

$$[\vec{r} - \vec{r}_0; \vec{a}] = R|\vec{a}|$$

10.28



$\vec{r}(\vec{r}_0)$ - конус

$$\vec{r} = \vec{r}_0 + \vec{a}t - \text{ось}$$

α - угол между осью и образующей

$$|[\vec{r} - \vec{r}_0; \vec{a}]| = |\vec{r} - \vec{r}_0| |\vec{a}| |\cos \alpha|$$

10.38

Написать ур-е плоского кругового цилиндра, проходящего через $\vec{r}(1;1;2)$ с осью $x=1+t, y=2+t, z=3+t$

$$\vec{a}(1;1;1) \quad \vec{r}_0(1;2;3) \quad \vec{r}_1(1;1;2) \quad \vec{r}(x;y;z)$$

$$[\vec{r}_1 - \vec{r}_0; \vec{a}] = \begin{vmatrix} i & j & k \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = (0; 1; 1)$$

$$|[\vec{r}_1 - \vec{r}_0; \vec{a}]| = \sqrt{2}$$

$$|\vec{a}| = \sqrt{3}$$

$$\Rightarrow R = \frac{\sqrt{2}}{\sqrt{3}}$$

$$[\vec{r} - \vec{r}_0; \vec{a}] = \begin{vmatrix} i & j & k \\ x-1 & y-2 & z-3 \\ 1 & 1 & 1 \end{vmatrix} = (y-z+1; -x+z-2; x-y+1)$$

$$(y-z+1)^2 + (-x+z-2)^2 + (x-y+1)^2 = \frac{2}{3} \cdot 3 = 2$$

10.39

$$\vec{a}(1;1;1) \quad \vec{r}_0(0;0;0) \quad \alpha = \arccos \frac{1}{\sqrt{3}}$$

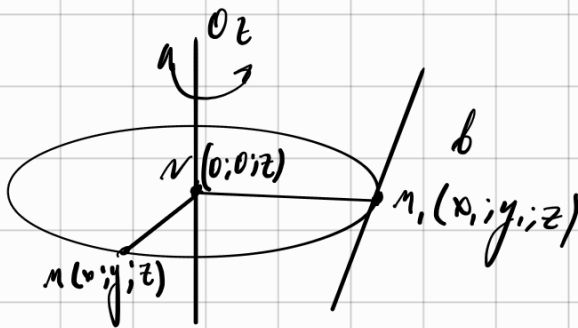
$$\vec{r}(x;y;z)$$

$$(x+y+z)^2 = (x^2+y^2+z^2) \cdot \frac{1}{\sqrt{3}}$$

$$x^2 + y^2 + z^2 + 2xy + 2xz + 2yz = x^2 + y^2 + z^2$$

$$xy + yz + zx = 0$$

10.40 Найми y -е нон. мун ноб-мн баярлан уяран $x=1+t$
 баярлан оон Oz $y=z=3+t$



$$|NM| = \sqrt{x^2 + y^2 + z^2}$$

$$NM^2 = x^2 + y^2 + z^2$$

$$x^2 + y^2 + z^2 = x^2 + y^2 + z^2$$

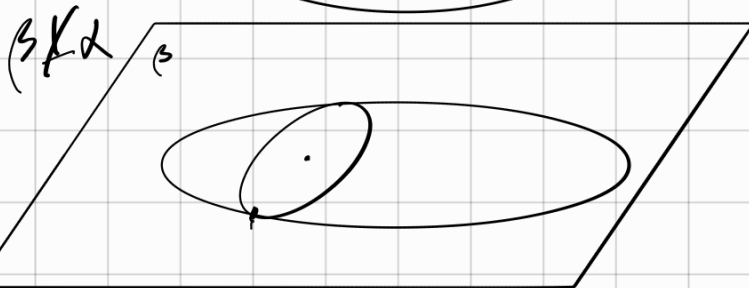
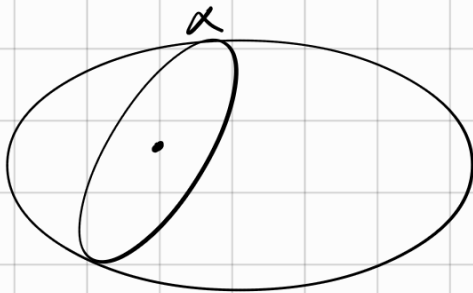
by y -а уяран $y=z$ $x=1+t$
 $(x=1+t \quad y=3+t \quad z=3+t)$

$$x^2 + y^2 = (z-1)^2 + z^2$$

$$x^2 + y^2 = z^2 - 4z + 4 + z^2 = 2(z^2 - 2z + 1) + 2$$

$$x^2 + y^2 = 2(z-1)^2 + 2 \quad ; \quad \frac{x^2}{2} + \frac{y^2}{2} - (z-1)^2 = 1 \Rightarrow \text{огном. мун.}$$

10.65 (2) Найми улам сөрөн $x^2 + 2y^2 + 4z^2 = 40$ $11-10$ $x+y+z=z$



$$x = z - y - z \quad \text{б ур-е } \exists 1-11:$$

$$(z-y-z)^2 + 2y^2 + 4z^2 - 40 = 0$$

$$4y - 14y - 14z + 2yz + y^2 + z^2 + 4z^2 + 4z^2 - 40 = 0$$

$$3y^2 + 2yz + 5z^2 - 14y - 14z + 7 = 0$$

A 2B C 2B 2E

$$\begin{cases} Ay_0 + Bz_0 + D = 0 \\ By_0 + Cz_0 + E = 0 \end{cases}$$

$$\begin{cases} 3y_0 + z_0 = 7 \\ y_0 + 5z_0 = 7 \\ y_0 = 2 \quad z_0 = 1 \\ x_0 = 4 \end{cases}$$

10.81 Найдем уравнение осей. $4x^2 - y^2 = 16z$ похоры xy $\gamma(2; 0; 1)$

Ответ: $(4; 2; 1)$

$$\begin{cases} x = 2 + a_1 t \\ y = a_2 t \\ z = 1 + a_3 t \end{cases} \quad 4(2 + a_1 t)^2 - (a_2 t)^2 = 16 + 16a_3 t$$

$$\begin{aligned} 4(2 + a_1 t)^2 - (a_2 t)^2 - 16(1 + a_3 t) &= 0 \\ 16 + 16a_1 t + 4a_1^2 t^2 - a_2^2 t^2 - 16 - 16a_3 t &= 0 \\ (4a_1^2 - a_2^2)t^2 + 16(a_1 - a_3)t &= 0 \end{aligned}$$

$$4a_1^2 = a_2^2$$

$$a_1 = a_3$$

$$a_1 = 1$$

$$a_3 = 1$$

$$a_2 = \pm 2$$

Ответ: $\begin{cases} x = 2 + t \\ y = \pm 2t \\ z = 1 + t \end{cases}$

$$4x^2 - y^2 = 16z$$

$$(2x - y)(2x + y) = 16z$$

$$\begin{aligned} \textcircled{1} \begin{cases} p(2x - y) = z \\ 2x + y = 16p \end{cases} & \quad \textcircled{2} \begin{cases} 2x - y = qz \\ q(2x + y) = 16 \end{cases} \end{aligned}$$

$$1) \begin{cases} 4p = 1 \\ 4 = 16p \end{cases} \quad p = 1/4 \quad \begin{cases} 2x - y - 4z = 0 \\ 2x + y - 4 = 0 \end{cases}$$

$$\textcircled{2} \quad q = 4 \quad \begin{cases} 2x + y - 4z = 0 \\ 2x - 4y - 16 = 0 \end{cases}$$

12.28 1) f - app. η -e

$$f(A)=A \quad f(B)=B$$

(!) AB - garis lurus monoton $\forall c \in AB \quad f(c) \in c$



$$\text{Dgr. } \vec{AB} \parallel \vec{CD}$$

$$\vec{CD} \neq 0 \quad \frac{\vec{AB}}{\vec{CD}} = \lambda$$

$$\text{ye } \lambda = \frac{|\vec{AB}|}{|\vec{CD}|}, \text{ yu } \vec{AB} \parallel \vec{CD}$$

$$\lambda = -\frac{|\vec{AB}|}{|\vec{CD}|}, \text{ yu } \vec{AB} \parallel \vec{CD}$$

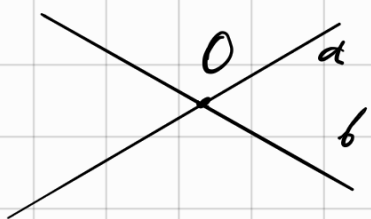
$$\forall c \text{ emu } \frac{\vec{AC}}{\vec{BC}} = \lambda,$$

mo mo cb-by app. η -g:

$$\frac{\vec{AC}^*}{\vec{BC}^*} = \lambda \Rightarrow \frac{\vec{AC}^*}{\vec{BC}^*} = \lambda = \frac{\vec{AC}}{\vec{BC}} \Rightarrow C = C^* = f(c)$$

$$3) f(a)=a \quad f(b)=b \quad a \cap b = 0$$

$$(1) f(0)=0$$

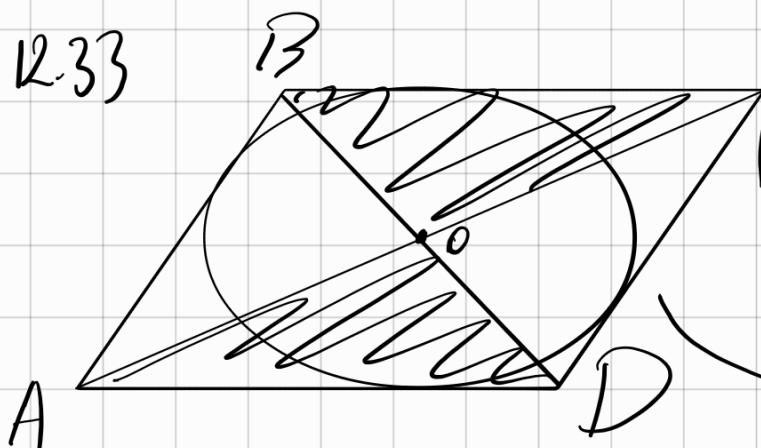


$$0 \in a \Rightarrow f(0) \in f(a)=a$$

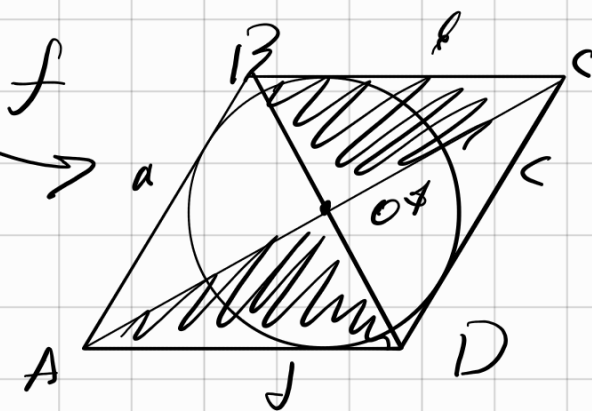
$$0 \in b \Rightarrow f(0) \in f(b)=b \Rightarrow$$

$$\Rightarrow f(0) = f(a) \cap f(b) = a \cap b = 0$$

12.33



$$(1) S_{\square} = S_{\square}$$



$$a + c = b + d$$

$$S_{\square} = \frac{1}{2}(a+c)R = S_{\square} = \frac{1}{2}(b+d)R$$

9.13 (1, 4)

$$1) (3x - 4y)^2 - 5(x + 2y - 1)^2 = 1$$

$$\begin{cases} x^* = 3x - 4y \\ y^* = x + 2y - 1 \end{cases}$$

$$\Delta = \begin{vmatrix} 3 & -4 \\ 1 & 2 \end{vmatrix} \neq 0 \Rightarrow \text{аффинн.}$$

$$x^{*2} - 5y^{*2} = 1 \quad \nwarrow \text{гипербола}$$

$$4) (4x + 3y - 1)^2 + (4x + 3y + 2)^2 = 5$$

$$\begin{cases} x^* = 4x + 3y - 1 \\ y^* = 4x + 3y + 2 \end{cases}$$

$$\Delta = \begin{vmatrix} 4 & 3 \\ 4 & 3 \end{vmatrix} = 0 \Rightarrow \text{не аффинное}$$

$$t = 4x + 3y$$

$$(t - 1)^2 + (t + 2)^2 = 5$$

$$t^2 - 2t + 1 + t^2 + 4t + 4 = 5$$

$$2t^2 + 2t - 0$$

$$t(t + 1) = 0$$

$$(4x + 3y)(4x + 3y + 1) = 0$$

две паралл. прямые

$$12.43(1) \begin{cases} x^* = 2x + 3y \\ y^* = -y \end{cases}$$

$$A = \begin{pmatrix} 2 & 3 \\ 0 & -1 \end{pmatrix}$$

$$\vec{x}^* = A \vec{x}$$

$$\det(A - \lambda E) = 0$$

$$\begin{vmatrix} 2 - \lambda & 3 \\ 0 & -1 - \lambda \end{vmatrix} = -(\lambda - 2)(\lambda + 1) = 0$$

$$A_{\lambda} = (A - \lambda I) \quad \lambda_1 = 2 \quad \lambda_2 = -1$$

$$\lambda = 2 \quad \begin{pmatrix} 0 & 3 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad h_i = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -B \\ A \end{pmatrix}$$

$$\lambda = -1 \quad \begin{pmatrix} 3 & 3 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad h_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

inv yzale:

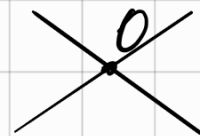
$$Ax + By + C = 0$$

$$1) -y + C_1 = 0$$

$$\begin{pmatrix} -B \\ A \end{pmatrix} - \text{rang}$$

$$c) -x - y + C_2 = 0$$

$$(0,0) \rightarrow (0,0)$$



12.28 (3)

Наименование работы

12.25/6) $\nu(r)$ $\rightarrow$ $\nu = \nu_0 + a t$
с коэффициентом $\lambda \geq 0$

$$\vec{n^+N} = 1 \vec{nN}$$

$$\vec{r}_N = \vec{r}_0 + \frac{(\vec{r} - \vec{r}_0, \vec{a})}{|\vec{a}|^2} \vec{a}$$

$$\vec{r}^{\rightarrow} = \lambda \vec{r} + (1-\lambda) \vec{r}_N =$$

$$= \lambda \vec{r} + (1-\lambda) \vec{r}_0 + (1-\lambda) \frac{(\vec{r}-\vec{r}_0, \vec{a})}{|\vec{a}|^2} \vec{a}$$

12.53(8) Cramme K $x+y-2=0$ $\lambda = \frac{1}{3}$

$$\vec{v}_0(1;1) \quad \vec{a}(-1;1) \quad (\vec{r}-\vec{r}_0, \vec{a}) = (x-1)(-1) + (y-1)(1) = y-x$$

$$x^{\Phi} = \frac{1}{3}x + \frac{2}{3} + \frac{2}{3} \frac{(y-x)}{2}(-1)$$

$$x^{\Phi} = \frac{1}{3}x + \frac{1}{3}x - \frac{1}{3}y + \frac{2}{3} = \frac{2}{3}x - \frac{1}{3}y + \frac{2}{3}$$

$$y^{\Phi} = \frac{1}{3}y + \frac{2}{3} \frac{(y-x)}{2} \cdot 1 + \frac{2}{3}$$

$$y^{\Phi} = \frac{2}{3}y - \frac{1}{3}x + \frac{2}{3}$$

12.82(8) Найти новое разложение апп у-я

$$\begin{cases} x^{\Phi} = -4x + 7y \\ y^{\Phi} = 8x + y \end{cases}$$

1) Найти новые канонич.

$$\vec{p} = \begin{pmatrix} \xi \\ \eta \end{pmatrix} \quad \vec{q} = \begin{pmatrix} \eta \\ -\xi \end{pmatrix} \quad \text{где max, чтобы: } (\vec{p}; \vec{q}) = 0$$

$$\begin{pmatrix} -4\xi + 7\eta \\ 8\xi + \eta \end{pmatrix} = \vec{p}^{\Phi} \quad \vec{q}^{\Phi} = \begin{pmatrix} -4\eta - 7\xi \\ 8\eta - \xi \end{pmatrix} \quad (\vec{p}^{\Phi}; \vec{q}^{\Phi}) = 0$$

$$\begin{cases} \xi^{\Phi} = -4\xi + 7\eta \\ \eta^{\Phi} = 8\xi + \eta \end{cases}$$

$$(-4\xi + 7\eta)(-4\eta - 7\xi) + (8\xi + \eta)(8\eta - \xi) =$$

$$= 16\xi\eta - 28\eta^2 - 28\xi^2 - 49\xi\eta + 64\xi\eta + 8\eta^2 - 8\xi - \xi\eta =$$

$$= 20\xi^2 + 30\xi\eta - 20\eta^2 = 0$$

$$2\xi^2 + 3\xi\eta - 2\eta^2 = 0$$

$$2\left(\frac{\xi}{\eta}\right)^2 + 3\frac{\xi}{\eta} - 2 = 0$$

$$\frac{\xi}{\eta} = -2$$

$$\frac{\xi}{\eta} = \frac{1}{2}$$

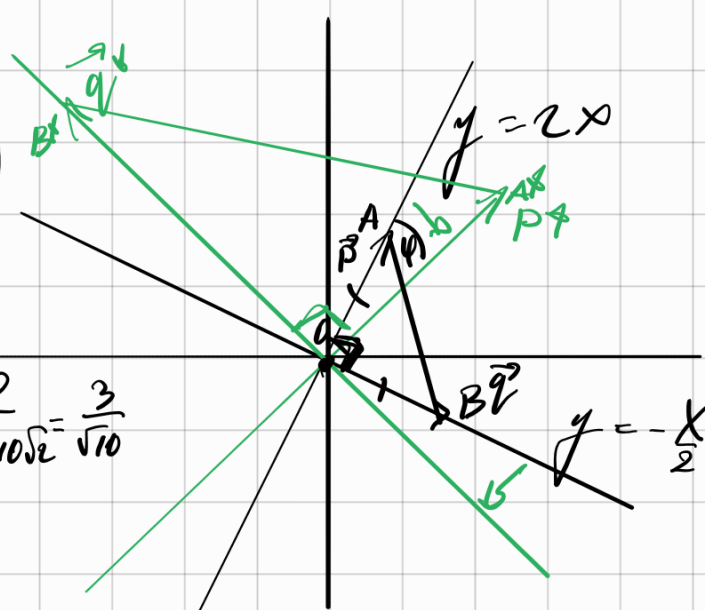
Возьмем $\xi = 1$ $\eta = 2$

$$P = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad q = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$P^* = \begin{pmatrix} 10 \\ 10 \end{pmatrix} \quad q^* = \begin{pmatrix} -15 \\ 15 \end{pmatrix}$$

$$\cos \varphi = \frac{(P, P^*)}{|P| |P^*|} = \frac{30}{\sqrt{5} \cdot 10\sqrt{2}} = \frac{3}{\sqrt{10}}$$

$$\varphi = \arccos \frac{3}{\sqrt{10}}$$



2) Определим φ : $\varphi = S_e R_0^{-1}$, где $O(0;0)$ $\varphi = \arccos \frac{3}{\sqrt{10}}$

3) h_1 : симметрия к $y = -x$ с коэф $\lambda_1 = \frac{|P^*|}{|P|} = 2\sqrt{10}$

h_2 : симметрия к $y = x$ с коэф $\lambda_2 = \frac{|q^*|}{|q|} = 3\sqrt{10}$

18 Dec

13:55

426

ГК

14.12

1) Если мы в φ -е зададим $\det A_5$ значение

$$\ominus a_{15} a_{12} a_{34} a_{21} a_{43}$$

$$a_{55} a_{12} a_{34} a_{21} a_{43}$$

$$\oplus a_{12} a_{21} a_{34} a_{43} a_{55}$$

2) С каких знаков в det A_5 берем $a_{12} a_{21} a_{34} a_{43} a_{55}$ и

$$\pi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 5 & 3 \end{pmatrix} \quad \pi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 4 & 1 & 2 \end{pmatrix}$$

$$\text{inv } \pi_1 = 1 + 0 + 1 + 1 + 0 = 3 \quad \text{inv } \pi_2 = 4 + 2 + 2 = 8$$

$$\text{Знак } (-1)^3 = -1 \quad (-1)^8 = 1$$

14.14 $\Delta_n = \begin{vmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{vmatrix} = \lambda_1 \begin{vmatrix} \lambda_2 & 0 & \dots & 0 \\ 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{vmatrix} = \dots = \lambda_1 \lambda_2 \dots \lambda_n$

14.15 $\Delta_n = \begin{vmatrix} \lambda_1 & \lambda_2 & \dots & \lambda_n \\ 0 & \lambda_2 & \dots & \lambda_n \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{vmatrix} = \lambda_1 \begin{vmatrix} \lambda_2 & \dots & \lambda_n \\ 0 & \dots & \lambda_n \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{vmatrix} = \dots = \lambda_1 \lambda_2 \dots \lambda_n$

14.23(b) $\Delta_n = \begin{vmatrix} 0 & \dots & \lambda_1 \\ \lambda_n & \dots & 0 \end{vmatrix} = (-1)^n \begin{vmatrix} \lambda_1 & 0 & \dots & \lambda_n \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{vmatrix} = \lambda_1 \lambda_2 \dots \lambda_n (-1)^{n+n-1} = \dots =$

$$= \lambda_1 \lambda_2 \dots \lambda_n (-1)^{\frac{n(n+1)}{2}}$$

НАДО СКО $\frac{n(n-1)}{2}$, ТАМ нечетное количество строк

14.23(11)

$$\Delta_n = \begin{vmatrix} 1 & 2 & \dots & 22 \\ 2 & 1 & \dots & 22 \\ \vdots & \vdots & \ddots & \vdots \\ 2 & 2 & \dots & 12 \\ 2 & 2 & \dots & 21 \end{vmatrix} \xrightarrow{\text{избавимся от 2-х строк, вычтем 1, вычтем 1}} \begin{vmatrix} 1 & 2 & \dots & 22 \\ 1 & -1 & \dots & 0 \\ 1 & 0 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & -1 \end{vmatrix} \xrightarrow{\text{избавимся от 1-х строк, вычтем 1, вычтем 1}} \begin{vmatrix} 1 & 2 & \dots & 22 \\ 0 & -2 & \dots & -22 \\ 0 & -1 & \dots & -22 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & -1 & \dots & -22 \end{vmatrix}$$

$$= \begin{vmatrix} (1+2(n-1)) & 0 & \dots & 0 \\ 1 & -1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \end{vmatrix} \xrightarrow{\text{разложим по 1-й строке}} (2n-1) \begin{vmatrix} 1 & 0 & \dots & 0 \\ -1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & -1 & \dots & -1 \end{vmatrix} = (2n-1) (-1)^{n-1}$$

14.23(16)

$$\Delta_n = \begin{vmatrix} 2 & 1 & \dots & 0 \\ 1 & 2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 2 \end{vmatrix} \xrightarrow{\text{разложим по 1-й строке}} 2\Delta_{n-1} - \begin{vmatrix} 1 & 1 & \dots & 0 \\ 0 & 2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 2 \end{vmatrix} = 2\Delta_{n-1} - \Delta_{n-2}$$

$$\Delta_n = 2\Delta_{n-1} - \Delta_{n-2} \quad n \geq 3$$

$$\Delta_1 = 2$$

$$\Delta_2 = 3$$

$$\Delta_3 = 6 - 2 = 4$$

$$\Delta_n + \Delta_{n-1} + \dots + \Delta_3 = 2\Delta_{n-1} + 2\Delta_{n-2} + \dots + 2\Delta_2 - \Delta_{n-2} - \Delta_{n-3} - \Delta_1$$

$$+ \begin{cases} \Delta_n = \Delta_{n-1} + \Delta_2 - \Delta_1 = \Delta_{n-1} + 1 \\ \Delta_{n-1} = \Delta_{n-2} + 1 \\ \vdots \\ \Delta_2 = \Delta_1 + 1 \end{cases}$$

$$\Delta_n = \Delta_1 + (n-1) = n+1$$

$$\Delta_n = a\Delta_{n-1} + b\Delta_{n-2} \quad \Delta_1, \Delta_2 - \text{uzbekmas}$$

$$q^n = aq^{n-1} + bq^{n-2}$$

$$q^2 - aq + b = 0$$

$$1) q_1 \neq q_2 \quad \Delta_n = C_1 q_1^n + C_2 q_2^n$$

$$2) q_1 = q_2 = q \quad \Delta_n = (C_1^n + C_2) q^n$$

$$q^2 - 2q + 1 > 0 \quad q_{1,2} = 1$$

$$\Delta_n = (C_1 n + C_2) 1^n$$

$$C_1 + C_2 = 2$$

$$2C_1 + C_2 = 3$$

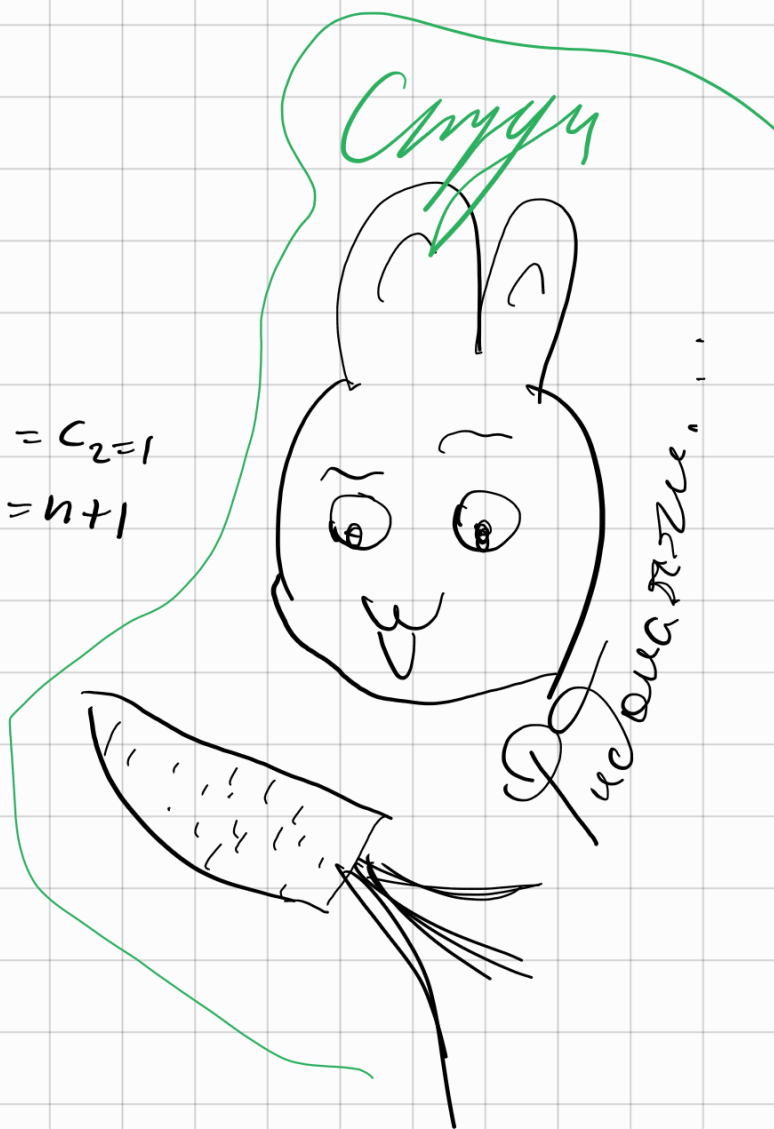
$$C_1 = C_2 = 1$$

$$\Delta_n = n+1$$

14.24(7) Вандермонд

$$\Delta_n = W(x_1, \dots, x_n) =$$

$$= \begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ x_1 & x_2 & x_3 & \dots & x_n \\ x_1^2 & x_2^2 & x_3^2 & \dots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \dots & \dots & x_n^{n-1} \end{vmatrix}$$



$$P(x) = \begin{vmatrix} 1 & x & \dots & x^{n-1} \\ x_1 & x_2 & \dots & x_{n-1} \\ x_1^2 & x_2^2 & \dots & x_{n-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \dots & x_{n-1}^{n-1} \end{vmatrix} - \text{множитель степени } n-1$$

корни: $x_1, x_2, \dots, x_{n-1}$

$$P(x) = C (x-x_1)(x-x_2) \dots (x-x_{n-1})$$

$$C = \Delta_{n-1}$$

$$\begin{cases} \Delta_n = \Delta_{n-1} (x_n - x_1)(x_n - x_2) \dots (x_n - x_{n-1}) \\ \Delta_{n-1} = \Delta_{n-2} (x_{n-1} - x_1)(x_{n-1} - x_2) \dots (x_{n-1} - x_{n-2}) \\ \vdots \\ \Delta_2 = \Delta_1 (x_2 - x_1) \end{cases}$$

$$\Delta_n = \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

15.11(7)

$$A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

Классу A^n ← n беремо $\text{rea } n\alpha$

$$A^n = \begin{pmatrix} \cos n\alpha & -\sin n\alpha \\ \sin n\alpha & \cos n\alpha \end{pmatrix}$$

15.22(2)

$$f(t) = t^2 - 2t + 1$$

$$\text{Классу } f(A)$$

$$f(A) = A^2 - 2A + E = (A - E)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$(A - B)^2 = A^2 - AB - BA + B^2 \quad B = E$$

$$X(t) = \begin{vmatrix} 1-t & 1 \\ 0 & 1-t \end{vmatrix} = (1-t)^2 = 0 \quad t=1$$

" $t^2 - 2t + 1$

$$\text{По т. Кан-Крем } K(A) = \underline{\underline{0}}$$

15.45(1)

$$A = \begin{pmatrix} 3 & 5 \\ 5 & 9 \end{pmatrix} \text{ найти } A^{-1}$$

$$1^{\text{й}} \text{ м. } \Delta A = 2 \neq 0 \Rightarrow \exists A^{-1}$$

$$x_{ij} = \frac{(-1)^{i+j} a_{ji}}{\Delta A}$$

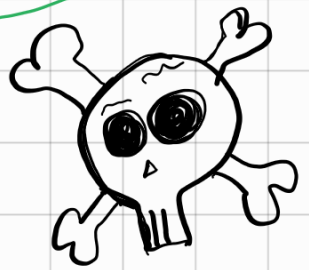
$$x_{11} = \frac{9}{2}$$

$$x_{12} = -\frac{5}{2}$$

$$x_{21} = -\frac{5}{2}$$

$$x_{22} = \frac{3}{2}$$

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 9 & -5 \\ -5 & 3 \end{pmatrix}$$



copy

$$2^{\text{й}} \text{ м. } \left(\begin{array}{cc|cc} 3 & 5 & 1 & 0 \\ 5 & 9 & 0 & 1 \end{array} \right) \xrightarrow[A \quad E]{} \left(\begin{array}{cc|cc} 1 & \frac{5}{3} & \frac{1}{3} & 0 \\ 5 & 9 & 0 & 1 \end{array} \right) \rightarrow$$

$$\rightarrow \left(\begin{array}{cc|cc} 1 & \frac{5}{3} & \frac{1}{3} & 0 \\ 0 & \frac{2}{3} & -\frac{5}{3} & 1 \end{array} \right) \rightarrow \left(\begin{array}{cc|cc} 1 & \frac{5}{3} & \frac{1}{3} & 0 \\ 0 & 1 & -\frac{5}{2} & \frac{3}{2} \end{array} \right) \rightarrow$$

$$\rightarrow \left(\begin{array}{cc|cc} 1 & 0 & \frac{9}{2} & -\frac{5}{2} \\ 0 & 1 & -\frac{5}{2} & \frac{3}{2} \end{array} \right) \xrightarrow[E \quad A^{-1}]{} \left(\begin{array}{cc|cc} 1 & 0 & \frac{9}{2} & -\frac{5}{2} \\ 0 & 1 & -\frac{5}{2} & \frac{3}{2} \end{array} \right)$$

15.45(5)

$$A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} = A^T$$

$$A^T A = A A^T = E \Leftrightarrow A - \text{ортонорм. матрица}$$

$$\Delta A^T A = 1 \quad (\Delta A)^2 = 1$$

$$\Delta A^T \cdot \Delta A = 1 \quad |\Delta A| = 1$$

$$\Delta A$$

15.55

$$A^2 + A + E = 0$$

$$A(A+E) = -E$$

$$\det A \det(A+E) = (-1)^n \neq 0$$

$$\det A \neq 0$$

$$A^{-1}(A^2 + A + E) = 0$$

$$A^{-1}AA + E + A^{-1} = 0$$

$$A^{-1} = -A - E$$

15.64 (5)

$$A(X+C) = B, \text{ let } A \neq 0 \Rightarrow \exists A^{-1}$$

$$X+C = A^{-1}B$$

$$X = A^{-1}B - C$$